G. Narayanamma Institute of Technology & Science

(Autonomous) (For women)

Shaikpet, Hyderabad - 500 104

II B.Tech I Semester – I MID (Special) Examinations February – 2023

Academic Year 2022-2023

SUBJECT: Digital Logic Design

SUBJECT CODE: ES123AN

BRANCH: CSE, CSM & CSD

Date: 22/02/23(A.N)

Max Marks: 25

Time: 2Hrs

Blooms Taxonomy Levels:
Level 1 – Remembering, Level 2 – Understanding, Level 3 – Applying, Level 4 – Analyzing, Level 5 – Evaluating, Level 6 - Creating

PART-A

(Answer all the questions. Each question carries 1 mark) **1x10=10 Marks**

S.No. 1	Question	CO	BT Level
a)	2's complement of 1010110 is _____	CO1	L-1
b)	Draw the symbol and truth table for OR gate.	CO1	L-1
c)	Mention the truth table for Associative Law.	CO1	L-2
d)	Find the 9's complement of $(7218)_{10}$.	CO1	L-2
e)	Name the universal gates and draw their symbols & truth tables.	CO2	L-1
f)	Define Maxterm.	CO2	L-1
g)	Write the advantages of K-Map.	CO2	L-1
h)	An n-variable K-Map can have _____ cells.	CO2	L-2
i)	What is combinational circuit?	CO3	L-1
j)	What is the purpose of the parity generator and checker?	CO3	L-1

21/2/23

PART-B

(Answer 03 full questions. Each question carries 5 marks) **3x5=15 Marks**

S.No.	Question	Marks	CO	BT Level
2.a)	Convert $(562)_{10}$ into the number with base 2, 8 and 16.	[2M]	CO1	L-3
b)	Write the steps to follow subtraction of two n-digit unsigned numbers M-N in base r. Compute X-Y with X=1010100 and Y=1000011 using 2's complement.	[3M]	CO1	L-2
3.a)	List out all postulates and theorems of Boolean Algebra.	[2M]	CO1	L-1
b)	Simplify the following Boolean expressions to a minimum number of literals (using theorems and postulates) i) ABC+A'B+ABC' ii) xy+x(wz+wz')	[3M]	CO1	L-3
4.a)	Write the steps to simplify Boolean function using 3 variable k-map.	[2M]	CO2	L-1
b)	Simplify the following functions using k-map method. i. $F(A, B, C) = \sum(2,3,6,7)$ ii. $F(w,x,y,z) = \pi(1,3,6,9,11,12,14)$	[3M]	CO2	L-3
5.	Simplify the following boolean function using four variable k-map method and illustrate the simplified function using NAND gates. $F(A,B,C,D) = \sum(0,2,4,5,6,7,8,10,13,15)$	[5M]	CO2	L-4
6.	Explain the design procedure for the combinational circuit and design a combinational circuit for converting BCD to Excess-3 code.	[5M]	CO3	L-3